Applying PML for 2D scalar waves

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1 Deriving the PML at frequency domain

1.1 Computational form of wave equation

Consider the source-free scalar wave equation:

$$\nabla \cdot (a\nabla u) = \frac{1}{b}\frac{\partial^2 u}{\partial t^2} \tag{1}$$

where $u(\mathbf{x}, t)$ is the scalar wave amplitude and $c = \sqrt{ab}$ is the phase velocity of the wave for some parameters $a(\mathbf{x})$ and $b(\mathbf{x})$ of the medium.

In order to solve this second-order-differential equation directly by the computing, instead we can separate the equation into two-coupled first-order differential equation, by using auxiliary field $\mathbf{v}(\mathbf{x}, t)$.

$$\frac{\partial u}{\partial t} = b\nabla \cdot \mathbf{v} \tag{2}$$

$$\frac{\partial \mathbf{v}}{\partial t} = a\nabla u \tag{3}$$

And we can stack these two equations into one linear differential equation by using matrix notation

$$\frac{\partial \mathbf{w}}{\partial t} = \frac{\partial}{\partial t}\begin{pmatrix} u \\ \mathbf{v} \end{pmatrix} = \begin{pmatrix} 0 & b\nabla\cdot \\ a\nabla & 0 \end{pmatrix}\begin{pmatrix} u \\ \mathbf{v} \end{pmatrix} = \hat{D}\mathbf{w} \tag{4}$$

where $\mathbf{w} = (u; \mathbf{v})$ is a stacked vector of $u(\mathbf{x}, t)$ and $\mathbf{v}(\mathbf{x}, t)$ and $\hat{D}$ is a linear operator of equation. The important point is that $\hat{D}$ is an anti-Hermitian operator in a proper choice of inner product. It's because taking a complex conjugate transpose occurs changes the form of operators ∇ to $-\nabla\cdot$ and $\nabla\cdot$ to $-\nabla$. So we can express $\hat{D}^H = -D$ and also we can ensure the eigenvalues of matrix are purely imaginary. By using this property, we can simplify the equation.

$$\frac{\partial \mathbf{w}}{\partial t} = \hat{D}\mathbf{w} = is\mathbf{w}, s \in R \tag{5}$$

So, by solving this equation we can express the oscillating solutions, and other wave-like phenomena.

$$\mathbf{w} = C(\mathbf{x})e^{ist} \tag{6}$$

1.2 Truncate the region using complex coordinate stretching

Let us start with the solution $\mathbf{w}(\mathbf{x}, t)$ of some wave equation in infinite space. Our purpose is to truncate space outside the region of interest in such way as to absorb radiating waves without reflection. This truncation occurs in three conceptual steps.

1. Analytically continue the solutions and equations to a complex x contour to make the waves exponentially decaying outside the region of the interest.
2. Perform a coordinate transformation to express the complex x as a function of a real coordinate.
3. Truncate the domain of this new real coordinate inside the complex-material region.

And also, to simplify the problem, let's assume that the region of interest is *homogeneous*, *linear* and *time-invariant*. Under these assumptions, the radiating solutions can be expressed as:

$$\mathbf{w}(\mathbf{x}, t) = \sum_{\mathbf{k},w} \mathbf{W}_{\mathbf{k},w} e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} \tag{7}$$

for some constant amplitudes $\mathbf{W}_{\mathbf{k},\omega}$, where ω is angular frequency and $\mathbf{k}$ is the wavevector which consist by the wavenumbers of each direction. Because the region is homogeneous and time invariant, we can ensure the amplitudes are constant, and also by linearity we can take the form of a superposition of planewaves.

1.2.1 Analytic continuation

First, we will consider about the waves propagating in the $+x$ direction. And then, use the same method to other direction. Since the wave propagates in the $+x$ direction, the y and z components lying in the plane perpendicular to the direction of propagation influence only the transverse structure of the wave. Using this property, the solution can be separated as follows.

$$\mathbf{W}(y, z)e^{i(kx-\omega t)} \tag{8}$$

In a homogeneous medium, the group velocity and the phase velocity generally have the same sign; therefore, we shall assume $k > 0$. By separating Eq. (8) into terms that depend on x and those that do not, we can easily found that it is an analytic function of x. It means that we can freely analytically continue the solution and equation to complex values of x.

Then, let's focus on the term e^{ikx}. By decomposing x into its real and imaginary parts, the solution can exponentially decay because $e^{ikx} = e^{ik(Rex+iImx)} = e^{ikRex}e^{-kImx}$ as Imx increases. So by using this property, if we increase the imaginary part of x only in the region outside the domain of interest, the solution of the governing equation remains unchanged within the domain of interest, and hence no reflection is generated at the interface. Meanwhile, in the exterior region, the wave is exponentially attenuated and effectively absorbed, causing the medium to behave as a so-called reflectionless absorbing material. This constitutes the fundamental principle of the perfectly matched layer (PML).

1.2.2 Coordinate transformation

Let's denote the complex x contour by $\tilde{x} = x + if(x)$, where $f(x)$ is the function how we've deformed our contour along the imaginary axis. It's hard to compute the complex coordinate, so instead, we will change variables to write equations in terms of x, the real part. Taking the derivative of $\tilde{x}$ respect to x, result is $\frac{\partial\tilde{x}}{\partial x} = 1+i\frac{df}{dx}$, so the differential along the deformed contour

$\tilde{x}$ can be expressed as $\partial\tilde{x} = (1 + i\frac{df}{dx})\partial x$. For convenience, let's denote $\frac{df}{dx} = \frac{\sigma_x(x)}{\omega}$, for some function $\sigma_x(x)$. Finally we can invert the derivative of $\tilde{x}$ as:

$$\frac{\partial}{\partial\tilde{x}} = \frac{1}{1 + i\frac{\sigma_x(x)}{\omega}}\frac{\partial}{\partial x} \tag{9}$$

Consequently, employing the above relation to rewrite the equation in terms of x instead of $\tilde{x}$ allows all operations to be carried out on the real coordinate space. In the PML regions, where $\sigma_x(x) > 0$, our oscillating solutions turn into exponentially decaying ones. In the interest regions, where $\sigma_x(x) = 0$, our wave equation is unchanged and the solution is also unchanged, so it does not requires numerical reflections.

An interesting aspects of the formulation is that $\frac{df}{dx}$ is defined as $\frac{\sigma_x(x)}{\omega}$ instead of simply $\sigma_x(x)$. The reason for choosing $\frac{\sigma_x(x)}{\omega}$ is to make the attenuation rate in the PML independent of the frequency ω. In the new coordinate system, we get:

$$e^{ik\tilde{x}} = e^{ikx}e^{-\frac{k}{\omega}\int^x \sigma_x(x')dx'} \tag{10}$$

Notice the factor $\frac{k}{w}$ is equal to $\frac{1}{c_x}$, the inverse of the phase velocity of x direction. In a dispersionless material, c_x is a constant independent of velocity for a fixed angle, in which case the attenuation rate in the PML is independent of frequency, and all wavelengths would decay at the same rate.

By the way, this may be regarded as the complexity associated with the complex coordinate transformation being mapped onto the material properties of the medium. Let us consider the governing equation that we previously derived. Applying the PML transformation to this equation at frequency domain, it is rewritten as follows:

$$-i\omega u = b\nabla \cdot \mathbf{v} = b(\frac{\partial v_x}{\partial\tilde{x}} + \frac{\partial v_y}{\partial\tilde{y}} + \frac{\partial v_z}{\partial\tilde{z}}) \tag{11}$$

$$-i\omega \mathbf{v} = a(\frac{\partial u}{\partial\tilde{x}}, \frac{\partial u}{\partial\tilde{y}}, \frac{\partial u}{\partial\tilde{z}}) \tag{12}$$

If the stretch factors along each coordinate direction are defined as s_x, s_y, s_z, respectively, By transferring the properties of this complex quantity to a and b, the equation can be expressed in an alternative form. For Eq. (11) :

$$-i\omega u = b(\frac{1}{s_x}\frac{\partial v_x}{\partial x} + \frac{1}{s_y}\frac{\partial v_y}{\partial y} + \frac{1}{s_z}\frac{\partial v_z}{\partial z}) = (\frac{b}{s_x}\frac{\partial v_x}{\partial x} + \frac{b}{s_y}\frac{\partial v_y}{\partial y} + \frac{b}{s_z}\frac{\partial v_z}{\partial z}) \tag{13}$$

From Eq. (13), the final term can be expressed as the inner product of the following vectors.

$$\frac{b}{s_x}\frac{\partial v_x}{\partial x} + \frac{b}{s_y}\frac{\partial v_y}{\partial y} + \frac{b}{s_z}\frac{\partial v_z}{\partial z} = (\frac{b}{s_x}, \frac{b}{s_y}, \frac{b}{s_z})\begin{pmatrix}\frac{\partial v_x}{\partial x}\\ \frac{\partial v_y}{\partial y}\\ \frac{\partial v_z}{\partial z}\end{pmatrix} = (b, b, b)\begin{pmatrix}\frac{1}{s_x} & & \\ & \frac{1}{s_y} & \\ & & \frac{1}{s_z}\end{pmatrix}\begin{pmatrix}\frac{\partial v_x}{\partial x}\\ \frac{\partial v_y}{\partial y}\\ \frac{\partial v_z}{\partial z}\end{pmatrix} \tag{14}$$

If the matrix $\mathbf{S}$ is defined as $diag(s_x, s_y, s_z)$, and $\mathbf{b}$ is defined as $\begin{pmatrix}b\\ b\\ b\end{pmatrix}$, we can express the equation as:

$$-iwu = \mathbf{b}^T\mathbf{S}^{-1}\nabla \cdot \mathbf{v} \tag{15}$$

Eq. (12) can also be transformed using the same approach. As the time derivative of $\mathbf{v}$ is a vector, by setting $A = diag(a, a, a)$, we can express the equation as:

$$-iw\mathbf{v} = \mathbf{S}^{-1} A \nabla u \tag{16}$$

Rewriting this transformation in the form of the original governing equation, it can be expressed as follows.

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ \mathbf{v} \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{b}^T \mathbf{S}^{-1} \nabla \\ \mathbf{S}^{-1}\mathbf{A}\nabla & 0 \end{pmatrix}\begin{pmatrix} u \\ \mathbf{v} \end{pmatrix} = \hat{D}_{pml}\begin{pmatrix} u \\ \mathbf{v} \end{pmatrix} \tag{17}$$

Expressed using index notation, this can be written as follows.

$$\partial_t \mathbf{w}_\mu = (\hat{D}_{pml})_{\mu\nu}\mathbf{w}_\nu \tag{18}$$

$$(\hat{D}_{PML})_{\mu\nu} = \begin{cases} \dfrac{b}{s_\nu}\partial_\nu(\cdot) & \text{if } \mu = 0, \nu \in \{1,2,3\} \quad \text{(Row 0: Div)} \\[2ex] \dfrac{a}{s_\mu}\partial_\mu & \text{if } \mu \in \{1,2,3\}, \nu = 0 \quad \text{(Col 0: Grad)} \\[2ex] 0 & \text{otherwise} \end{cases}$$

Since both $\hat{D}$ and $\hat{D}_{pml}$ are rank-2 tensors, the relationship between them can be mapped using a rank-4 tensor and expressed as follows:

$$(\hat{D}_{pml})_{\mu\nu} = S_{\mu\nu\alpha\beta}\hat{D}_{\alpha\beta} \tag{19}$$

By comparing each corresponding entry of the two matrices on a one-to-one basis, we can derive the values that $\mathcal{S}$ must take.

- $(\hat{D})_{\mu\nu} = \begin{cases} b\,\partial_\nu & (\mu = 0,\ \nu = k) \quad \text{Div} \\ a\,\partial_\mu & (\mu = k,\ \nu = 0) \quad \text{Grad} \\ 0 & \text{others} \end{cases}$
- $(\hat{D}_{PML})_{\mu\nu} = \begin{cases} \frac{1}{s_\nu}(b\partial_\nu) & (\mu = 0, \nu = k) \\ \frac{1}{s_\mu}(a\partial_\mu) & (\mu = k, \nu = 0) \\ 0 & \text{others} \end{cases}$

Comparing the two expressions above, the structure (i.e., the positions of the entries) of the matrix remains unchanged, and only the values at their respective positions are multiplied by a specific coefficient $1/s$. That is, the value must be zero when the input indices (α, β) and the output indices (μ, ν) are different, and it is nonzero only when they coincide. Using two Kronecker deltas, $\mathcal{S}$ can be represented as follows.

$$\mathcal{S}_{\mu\nu\alpha\beta} = \delta_{\mu\alpha}\delta_{\nu\beta} \times \Lambda_{\mu\nu} \tag{20}$$

$\Lambda_{\mu\nu}$ is a scaling factor between $\hat{D}_{pml}$ and $\hat{D}$.

In the Divergence region where $(\mu = 0, \nu = k)$, $b\partial_k$ becomes $\frac{1}{s_k} b\partial_k$, the scaling factor should be $\frac{1}{s_k}$. Also, In the Gradient region where $(\mu = k, \nu = 0)$, $a\partial_k$ becomes $\frac{1}{s_k} a\partial_k$, the scaling factor also should be $\frac{1}{s_k}$.

$$\mathcal{S}_{\mu\nu\alpha\beta} = \delta_{\mu\alpha}\delta_{\nu\beta} \begin{cases} \dfrac{1}{s_\nu} & \text{if } \mu = 0, \nu \in \{1,2,3\} \quad \text{(Scale Div by } 1/s_k) \\ \dfrac{1}{s_\mu} & \text{if } \mu \in \{1,2,3\}, \nu = 0 \quad \text{(Scale Grad by } 1/s_k) \\ 1 & \text{otherwise} \end{cases} \tag{21}$$

1.2.3 Truncating the computational region

Once we have performed the PML transformation of our wave equations, the solutions remain unchanged in our region of interest and exponentially decay in the outer regions. That means that we can truncate the computational region at some sufficiently large x, perhaps by putting a hard wall such as Dirichlet-Boundary Condition. This is possible because waves are attenuated by the PML, so that the field reaching the boundary is exponentially small. Even if such waves are reflected and propagate back toward the region of interest, they must traverse the PML again, and therefore their influence on the region of interest is negligible.

Then, how should we choose the thickness of the PML outside the region of interest? In principle, we can make the PML region as thin as we want, just by making σ_x larger. However, arbitrarily increasing σ_x can lead to excessively large attenuation within a single computational grid cell, which in turn may induce numerical reflections. So practically, we turn on $\sigma_x(x)$ quadratically or cubically from zero, over a region of length a half-wavelength or so.

2 Applying PML at the time domain

2.1 Transform from frequency domain to time domain

As we have seen, in frequency domain, we are solving for solutions for trivial case of ω so PML is almost trivial: we just apply the coordinate transformation to every $\frac{\partial}{\partial x}$ derivative in our wave equation. By the way, most numerical simulations, such as the FDTD method, are performed in the time domain.

But in the time domain, because of our complex stretch factor $(1+\frac{i\sigma}{w})$ is frequency-dependent, the problem becomes more complicated. As the time-domain wave function may superimpose multiple frequencies at once, we cannot interpret the exact value of ω, so it is hard to express a $\frac{1}{\omega}$ dependence in time-domain.

One possible approach to resolve this issue is to introduce a stretch factor based on a constant frequency ω_0. As ω_0 is typical , While waves with frequencies near ω_0 are well absorbed, the attenuation deteriorates for frequencies outside this band. Motivated by these limitations, we explored approaches to directly implement the $\frac{1}{\omega}$ dependence in the time domain. A representative method involves a mathematical formulation based on the Fourier transform. Multiplication by $-i\omega$ corresponds to differentiation in the time domain,therefore, we consider the stretch factor as $1-\frac{\sigma}{i\omega}$, from which it is straightforward to see that this stretch factor leads to integration in the time domain.

However, such an integral formulation introduces a convolution, necessitating the storage and use of field values at every past time step, which renders the approach nearly infeasible from a numerical standpoint. As an alternative, we adopt the Auxiliary Differential Equation (ADE) method, in which auxiliary field variables are introduced to eliminate the integral terms by reformulating them as time-differential equations.

2.2 PML for 2D scalar waves using ADE

From now on, we will derive the $+x$-direction PML for the 2D scalar wave equation. Since the problem is formulated in two dimensions, the governing equation is given by the following.

1. $\frac{\partial u}{\partial t} = b\nabla\cdot\mathbf{v} = b\frac{\partial v_x}{\partial x} + b\frac{\partial v_y}{\partial y} = -i\omega u$
2. $\frac{\partial v_x}{\partial t} = a\frac{\partial u}{\partial x} = -i\omega v_x$
3. $\frac{\partial v_y}{\partial t} = a\frac{\partial u}{\partial x} = -i\omega v_y$

Upon applying the PML transformation to each equation, Eq. (3) is unaffected since it describes the y-direction. Hence, we consider only Eqs. (1) and (2). Multiplying both sides by $1+\frac{i\sigma_x}{\omega}$ transform back to real x, we obtain:

$$b\frac{\partial v_x}{\partial x} + b\frac{\partial v_y}{\partial y}(1+\frac{i\sigma_x}{\omega}) = -i\omega u + \sigma_x u \tag{22}$$

$$a\frac{\partial u}{\partial x} = -i\omega v_x + \sigma_x v_x \tag{23}$$

Equation 12 is easy to transform back to time-domain because $-i\omega$ becomes a time derivative. Equation 11 however, is a term with an explicit $\frac{1}{\omega}$ factor. So let's set a new auxiliary field variable ψ, satisfying

$$-i\omega\psi = b\sigma_x\frac{\partial v_y}{\partial y} \tag{24}$$

to make the equation as the form of time-derivative of ψ. Now, we can Fourier transform everything back to the time-domain, to get a set of four time-domain equations with PML absorbing boundaries in the x -direction.

$$\frac{\partial u}{\partial t} = b\frac{\partial v_x}{\partial x} + b\frac{\partial v_y}{\partial y} + \psi - \sigma_x u \tag{25}$$

$$\frac{\partial v_x}{\partial t} = a\frac{\partial u}{\partial x} - \sigma_x v_x \tag{26}$$

$$\frac{\partial v_y}{\partial t} = a\frac{\partial u}{\partial y} \tag{27}$$

$$\frac{\partial \psi}{\partial t} = b\sigma_x\frac{\partial v_y}{\partial y} \tag{28}$$

where last equation for ψ is our auxiliary differential equation.